F78AA - Actuarial and Financial Mathematics A
Tutorial 1 – Interest rates and valuing single cashflows

Question 1

1. You invest £2 000 in a bank account on 1 January 2017. The bank pays interest at 4% per annum effective, compound.

   Calculate how much is in the bank account after (i) 1 year; (ii) 2 years; and (iii) 3 years.

2. Suppose the bank paid interest at 4% per annum effective, but *simple* interest instead of *compound*.

   Calculate much would have been in the account after 3 years.

Question 2 Calculate the following accumulations:

1. Accumulate £1 for 3 years at 5% per annum effective, giving your answer rounded to 6 decimal places.

2. Accumulate £10 000 for 5 years at 3% per half-year effective, giving your answer rounded to 2 decimal places.

3. Accumulate £700 for 18 months at 2.35% per quarter-year effective, giving your answer rounded to 2 decimal places. (Hint: how many months are in a quarter-year?)

Question 3 Calculate the present values on 1 January 2017 of the following payments at the given rates of interest:

1. £1 000 on 1 January 2018, at 7% per annum effective.

2. £10 000 on 1 July 2018, at 4% per quarter-year effective.

3. £18 000 on 1 October 2019, at 1% per month effective.

Give your answers rounded to 2 decimal places.

Question 4 Give your answers rounded to 6 decimal places in the following questions.

1. The effective rate of interest is 6.5% per annum. Calculate the effective rate of interest per month.

2. The effective rate of interest is 0.5% per month. Calculate the effective rate of interest per half-year.

3. The effective rate of interest is 6.5% per annum. Calculate the effective rate of discount per month.

4. The effective rate of discount is 0.5% per month. Calculate the effective rate of interest per annum.

Question 5 The effective rate of interest will be 4.5% per annum during 2017, 5.5% per annum during 2018 and 6.5% per annum thereafter. Round your answers to 2 decimal places, except where stated differently.

Draw a time-line to illustrate the information graphically and then calculate:

1. the accumulation of £1 000 from 1 January 2017 to 1 January 2021.
2. the accumulation of £8 000 from 1 July 2017 to 1 July 2019.
3. the accumulation *factor* from 1 January 2017 to 1 January 2021, with your answer rounded to 6 decimal places.
4. the present value at 1 January 2017 of £1 000 receivable on 1 January 2019.
5. the present value at 1 October 2017 of £1 000 receivable on 1 March 2020.

Question 6 Starting at 1 January 2017, the effective rate of interest will be 4% per half-year for 12 months, 3% per quarter-year for 15 months and 1% per month thereafter. Round your answers to 2 decimal places, except where stated differently.

Draw a time-line to illustrate the information graphically and then:

1. Calculate the accumulation factor from 1 January 2017 to 1 January 2020, rounded to 6 decimal places.
2. Calculate the accumulation of £15 000 from 1 May 2017 to 1 August 2019.
3. Calculate the present value at 1 January 2017 of £1 000 receivable on 1 January 2019.
4. Calculate the present value at 1 February 2017 of £10 receivable on 1 February 2020, rounded to 6 decimal places.
5. Calculate the discount factor from 1 February 2020 to 1 February 2017, rounded to 6 decimal places.

Solutions: Q1.1. (i) £2 080.00; (ii) £2 163.20; and (iii) £2 249.73. Q1.2 £2 240.00; Q2.1. £1.157625; Q2.2. £13 439.16; Q2.3. £804.68. Q3.1. £934.58; Q3.2. £7 903.15; Q3.3. £12 961.86. Q4.1. 0.005262; Q4.2. 0.030378; Q4.3. 0.005234; Q4.4. 0.061996. Q5.1. £1 250.45; Q5.2. £8 903.80; Q5.3. 1.250455; Q5.4. £907.05; Q5.5. £871.09. Q6.1. 1.371340; Q6.2. £19 066.63; Q6.3. £821.46; Q6.4. £7.267288; Q6.5. 0.726729.

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Solutions to Tutorial 1 – Interest rates and valuing single cashflows

Solution 6

1. $1.04^2 \times 1.03^5 \times 1.01^9 = 1.371340$ (to 6 d.p.)
2. £$15\,000 \times 1.04^{4/3} \times 1.03^5 \times 1.01^4 = 19\,066.63$ (to 2 d.p.).
3. £$1\,000 \times 1.03^{-4} \times 1.04^{-2} = 821.46$ (to 2 d.p.).
4. £$10 \times 1.01^{-10} \times 1.03^{-5} \times 1.04^{-11/6} = 7.267288$ (to 6 d.p.).
5. $1.01^{-10} \times 1.03^{-5} \times 1.04^{-11/6} = 0.726729$ (to 6 d.p.).